Augmentation and leakage

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Prepare the purpose, prerequisites and materials

> Prepare Augmentation and leakage: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Check whether a transformation preserves the task label

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Training a small image model

Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a complete image model with fixed features and one trained weight

Reference example, not the only acceptable wording: Teacher explanation: d = 2, h1 = 2 and h2 = 0. Score is 0.9, **prediction** class 1 and squared error (0.9 - 1) squared = 0.01.

Convolution as a local operation

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a local weighted sum at each valid image position

Reference example, not the only acceptable wording: Teacher explanation: Window rows are [2, 0] and [1, 3]. Products are 2, 0, 0 and 3, giving **output** 5.

Debugging shapes and gradients

Without opening the old lesson, explain one key idea from Debugging shapes and gradients. Give an example and a mistake that the idea helps you avoid.

Look for: Check dimension contracts before computing a layer

Reference example, not the only acceptable wording: Teacher explanation: No. Each weight row must have the same length as x.

Confirm the intended feature schema, then supply a valid two-feature **input** or deliberately redesign and retrain the layer; do not silently discard a value.

For the compact alternative schedule, use two 45-minute sessions. This chapter explicitly revisits Week 23: different pixel arrays do not guarantee independent **augmentation** evaluation.

Keep its original toy-domain claim narrow and use a new grouped plan for the new experiment. Use fictional source IDs rather than student photos. No new **training** framework is needed.

Outcomes

Check whether a transformation preserves the task label

Keep related examples in one split using **source group**s

Compare **augmentation** fairly without inflating independent **evidence**

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

Using final outcomes to choose parameters or settings makes that set part of model selection rather than unseen **evidence** for the selected model.

End goal for the pictorial case

By the end, I can explain **augmentation** and **leakage** using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Augmentation and leakage using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

Explain Augmentation and leakage · 1

Return to the binary image task: class 1 means the left column sum is strictly larger than the right column sum; equal sums belong to class 0. Let d be left sum minus right sum.

An **augmentation** is useful only under a stated task and transformation rule. More image files do not automatically mean more independent **evidence**.

Explain Augmentation and leakage · 2

Swapping rows preserves each column sum, so it preserves this task label for every valid image. Swapping columns changes d to -d. For nonzero d the class changes; for d = 0 it stays class 0.

Blindly copying the original label after every flip would create incorrect **training** targets.

A transformation may be valid for one task and invalid for another. A crop can remove the **evidence** needed for a label.

Choose allowed transformations using the task definition and **training evidence**, document the assumptions, and evaluate the choice on validation data.

This code assumes valid 2 by 2 binary lists. It constructs new lists instead of changing the **input**. The label function encodes the classroom task rule, not a general method for recovering unknown real-world labels.

If an allowed transformation changes a known target, the transformed example needs a justified new label. Do not invent labels for ambiguous cases.

Keep the source ID, transformation and resulting label with every derived image so a teacher can audit what happened.

Term	Meaning
Augmentation	Creating transformed training examples under a declared procedure
Label preservation	A transformation leaving the correct target unchanged for the stated task
Source group	An identifier joining records that must stay together when splitting
Leakage	Information crossing an evaluation boundary in a way that undermines the intended independent check

Every variant stays with its source group

SOURCE GROUP	ASSIGNED SPLIT	DERIVED RECORDS
A	Training	A original and A row swap
B	Validation	B original only
C	Final test	C original only

Assign source groups before generating training variants. A new filename does not make a derived image independent. Evaluation examples follow a fixed declared protocol, not random training augmentation.

Explain Augmentation and leakage · 3

Guided: the split is invalid for unseen-source evaluation because A2 derives from A regardless of filename. Keep the entire A group in **training** and replace the validation group, then rerun any affected selection.

Exit: budgets are 180 versus 60 updates, so they are not matched. Predeclare an equal update budget and record repetition policies; 36 records still represent only 12 original **source group**s.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Every variant stays with its source group

SOURCE GROUP	ASSIGNED SPLIT	DERIVED RECORDS
A	Training	A original and A row swap
B	Validation	B original only
C	Final test	C original only

Assign source groups before generating training variants. A new filename does not make a derived image independent. Evaluation examples follow a fixed declared protocol, not random training augmentation.

A training image has source ID A. Its row-swapped copy is called A2 and placed in validation. No filenames match across splits. Is the split valid for evaluating unseen source images? Describe the repair.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Augmentation and leakage

Keep every augmented image with its original **source group**.

The facts we will use

Supplied facts

Fictional sources A, B, C, D each have original+horizontal-flip copies. A/B train, C validation, D final test.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/24

Augmentation and leakage

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Track provenance

Each flipped copy retains its source ID and parent image reference.





2. Assign by source

A/B originals and flips give 4 training records; C gives 2 validation; D gives 2 final.





3. Check label meaning

Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

CHANGE ONE THING

Move D's flipped copy into training.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Track provenance	Each flipped copy retains its source ID and parent image reference.
2. Assign by source	A/B originals and flips give 4 training records; C gives 2 validation; D gives 2 final.
3. Check label meaning	Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign **source group**s to **training**, validation and final evaluation before producing copies.

Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

What this example does not establish

Keep the stated limit

Augmentation validity is task-specific, and it cannot manufacture new independent plants or repair inadequate sampling.

Change one thing in the pictured case

Change one thing

Change: Move D's flipped copy into **training**.

Predicted result

Prediction: Final-test group separation fails.

Why the result follows

Reason: The transformed D image is still derived from the held-out source, despite a new filename.

Ask while showing the picture

Ask while showing the picture · 4

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Augmentation and leakage, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve frozen evaluation and restate the strict left-column task label.
7 to 18 minutes	Trace row and column swaps, then complete W1 and W2 including the equality boundary.
18 to 32 minutes	Run or hand-trace the transformation code and inspect source preservation.
32 to 45 minutes	Use the visual and guided case to keep lineage groups together before augmentation.
Session 2 start to 8 minutes	Audit the Week 23 transformation relationships and complete W3.
8 to 20 minutes	Distinguish source counts from record counts and work through the update-budget example.
20 to 35 minutes	Collect W4 independently, requiring source counts, retained counts and exclusions.
35 to 45 minutes	Use W5 and the exit response to propose a fair comparison, then administer the fresh retry.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Use cards with source ID and variant ID in separate columns. Provide a split ledger and distinguish original count, variant count and total record count.

Extension: enumerate all sixteen binary images to verify row-swap preservation, count column-swap label changes, and identify the equality cases that defeat automatic label complementation.

Reduce help deliberately

Completion practice

Completion task: For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

Reasoning to listen for

Teacher check: Teacher explanation: Both differences are 0 and both labels are class 0. Complementing class 0 would give class 1, which is wrong for ties.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for **Augmentation** and **leakage**. Proposed explanation: “Every image flip preserves the label”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: Validity depends on the task; column swaps change non-tie labels here.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

How many independent **source group**s are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

One, D, with two related records. Count both **source group**s and image rows; never describe its flip as a new independent source. Changed-case check: Final-test group separation fails.

The transformed D image is still derived from the held-out source, despite a new filename.

Changed-case reasoning: Final-test group separation fails. The transformed D image is still derived from the held-out source, despite a new filename.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Run the investigation and diagnose misconceptions

> Run the Augmentation and leakage investigation and use the first unsupported step to choose a teaching response.

Every image flip preserves the label: Validity depends on the task; column swaps change non-tie labels here.

Different filenames prove independence: Lineage, shared subjects and identical content can connect different filenames.

More augmented files mean more independent examples: Variants retain dependence on their source; report both group and record counts.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Augmentation and leakage questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: the split is invalid for unseen-source evaluation because A2 derives from A regardless of filename. Keep the entire A group in **training** and replace the validation group, then rerun any affected selection.

Exit: budgets are 180 versus 60 updates, so they are not matched. Predeclare an equal update budget and record repetition policies; 36 records still represent only 12 original **source group**s.

W1 Check the label after a transformation

Workbook question · W1

For rows [0, 1], [0, 1], compute d and class before and after swapping rows and after swapping columns.

Teacher reasoning · W1

Teacher explanation: Original d = -2 and class 0. Row swapping leaves d = -2 and class 0. Column swapping gives rows [1, 0], [1, 0], d = 2 and class 1.

W2 Keep the equality boundary

Workbook question · W2

For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

Teacher reasoning · W2

Teacher explanation: Both differences are 0 and both labels are class 0. Complementing class 0 would give class 1, which is wrong for ties.

W3 Audit related records

Workbook question · W3

Training contains P original and Q original. Validation contains P crop and R original. Final test contains S original. P crop is derived from P original.

Identify the overlap and explain what must be rerun if this validation set already selected a model.

Teacher reasoning · W3

Teacher explanation: Source P crosses **training** and validation. Keep all P records in one split and choose distinct evaluation groups.

Rerun **training** and model selection on the corrected plan; the old validation score does not describe the repaired experiment.

W4 Design a grouped split independently

Workbook question · W4

Six fictional **source group**s J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to **training**, M to validation and N, O to final test.

Keep all four records per **training** group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

Teacher reasoning · W4

Teacher explanation: **Training**: 3 groups and 12 records. Validation: 1 group and 1 record. Final test: 2 groups and 2 records. Total retained records are 15. Nine derived evaluation records are excluded.

No group crosses splits; the small group counts still limit evaluation **evidence**.

W5 Match a training budget

Workbook question · W5

Eight source images each have an original plus two variants. Compare four epochs of these 24 records with four epochs of the eight originals, using one-example batches.

Give both update counts and one way to match budgets without using final outcomes.

Teacher reasoning · W5

Teacher explanation: Augmented run uses 96 updates; original-only run uses 32. Predeclare 96 updates for both, equivalent to 12 passes over the originals for the baseline.

Record repetition and **augmentation** policies, use validation for choices and keep final evaluation reserved.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires source counts 3, 1, 2 and retained counts 12, 1, 2, with 15 total records and 9 excluded derived records. Fresh retry: five groups A through E each have an original plus two variants.

Train on A and B using all records, validate on C original and test on D and E originals. Source counts are 2, 1, 2; retained records are 6, 1, 2; total 9, with 6 derived evaluation records excluded.

Three augmented epochs at batch size 1 use 18 updates, compared with 6 for three epochs over two originals.

Assess independent transfer

Independent question · withhold the key

Selected case: Six fictional **source group**s J, K, L, M, N and O each have one original and three derived records. Assign J, K and L to **training**, M to validation and N, O to final test.

Keep all four records per **training** group but only originals for evaluation. Give source and retained-record counts per split, total retained records and excluded derived records.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: **Training**: 3 groups and 12 records. Validation: 1 group and 1 record. Final test: 2 groups and 2 records. Total retained records are 15. Nine derived evaluation records are excluded.

No group crosses splits; the small group counts still limit evaluation **evidence**.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Augmentation and leakage to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled **augmentation** and **leakage evidence** sheet to Image-pattern Network Investigation. Show how you can check whether a transformation preserves the task label.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Use cards with source ID and variant ID in separate columns. Provide a split ledger and distinguish original count, variant count and total record count.

Extension: enumerate all sixteen binary images to verify row-swap preservation, count column-swap label changes, and identify the equality cases that defeat automatic label complementation.

Week 25 introduces sequence representations and the information preserved by order.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/25, ai-5/27, ai-5/30. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Augmentation and leakage lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Check whether a transformation preserves the task label

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: For rows [1, 0], [0, 1], compute d and class before and after a column swap. Why is always replacing a label by 1 minus the label incorrect?

Supported practice answer

Reasoned practice answer: Teacher explanation: Both differences are 0 and both labels are class 0. Complementing class 0 would give class 1, which is wrong for ties.

Check every learner before discussion: Use the worked case for **Augmentation** and **leakage**. Proposed explanation: “Every image flip preserves the label”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: Validity depends on the task; column swaps change non-tie labels here.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Move D’s flipped copy into **training**. Require a **prediction** and a reason before comparing. Expected result: Final-test group separation fails.

Reason: The transformed D image is still derived from the held-out source, despite a new filename.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the

new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

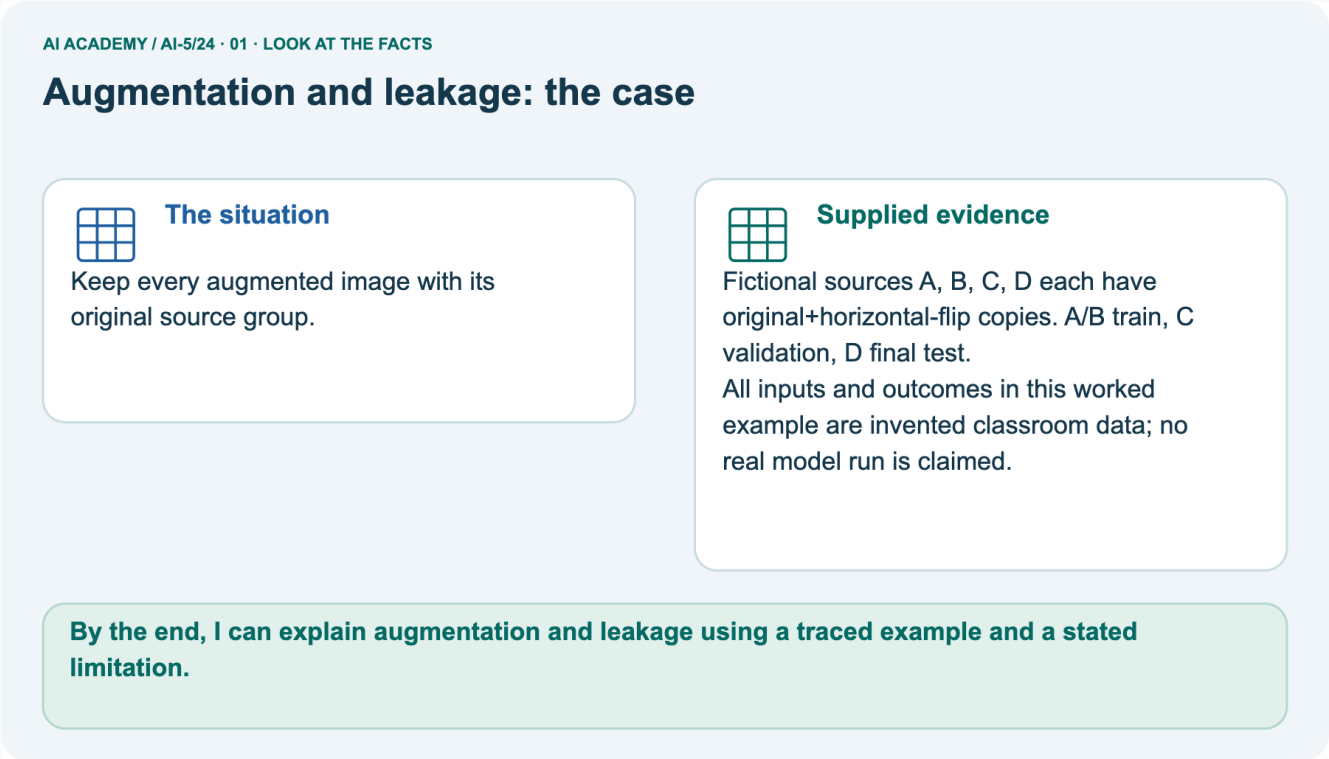
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Augmentation and leakage graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Augmentation and leakage: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Keep every augmented image with its original **source group**. Supplied facts: Fictional sources A, B, C, D each have original+horizontal-flip copies. A/B train, C validation, D final test.

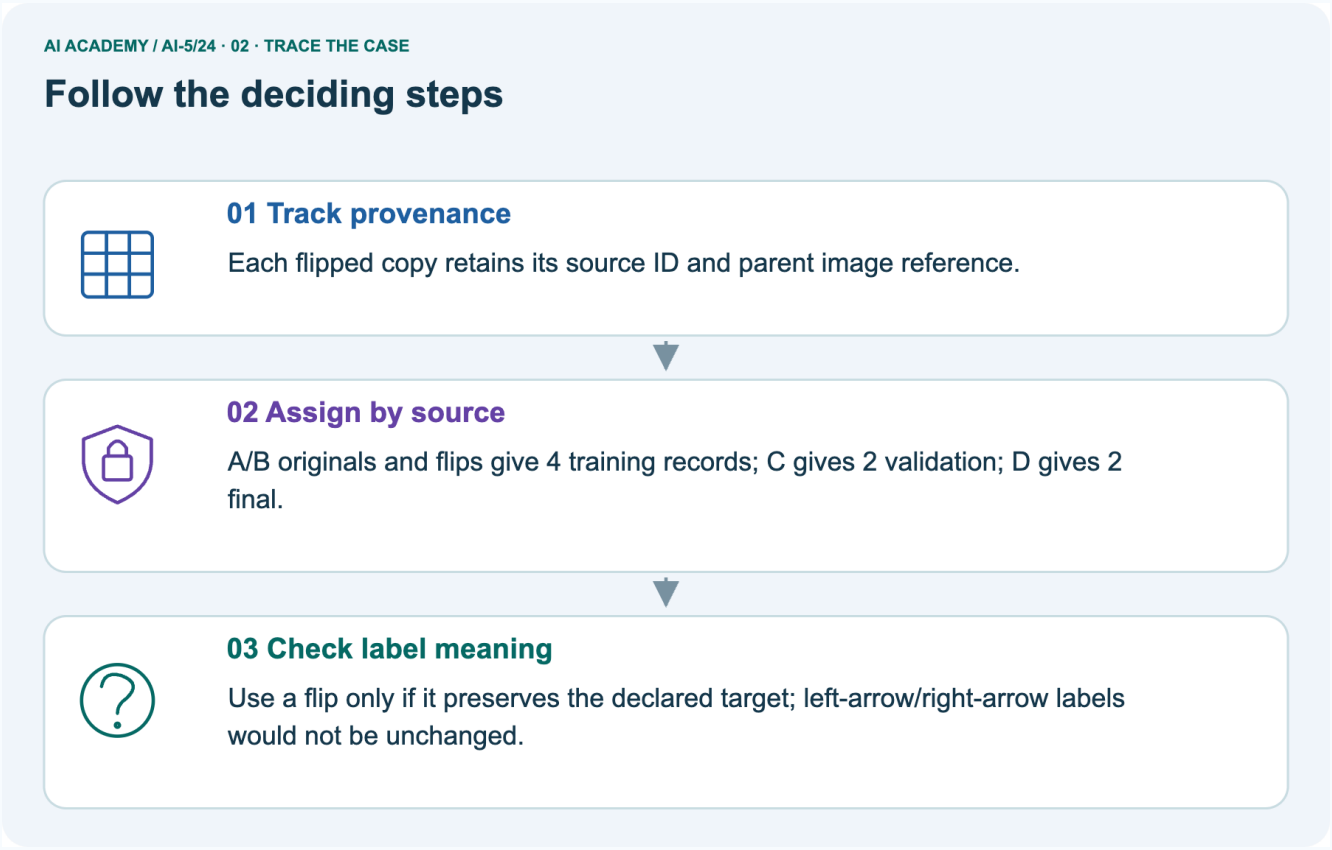
All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Track provenance: Each flipped copy retains its source ID and parent image reference. → Assign by source: A/B originals and flips give 4 **training** records; C gives 2 validation; D gives 2 final. → Check label meaning: Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged.

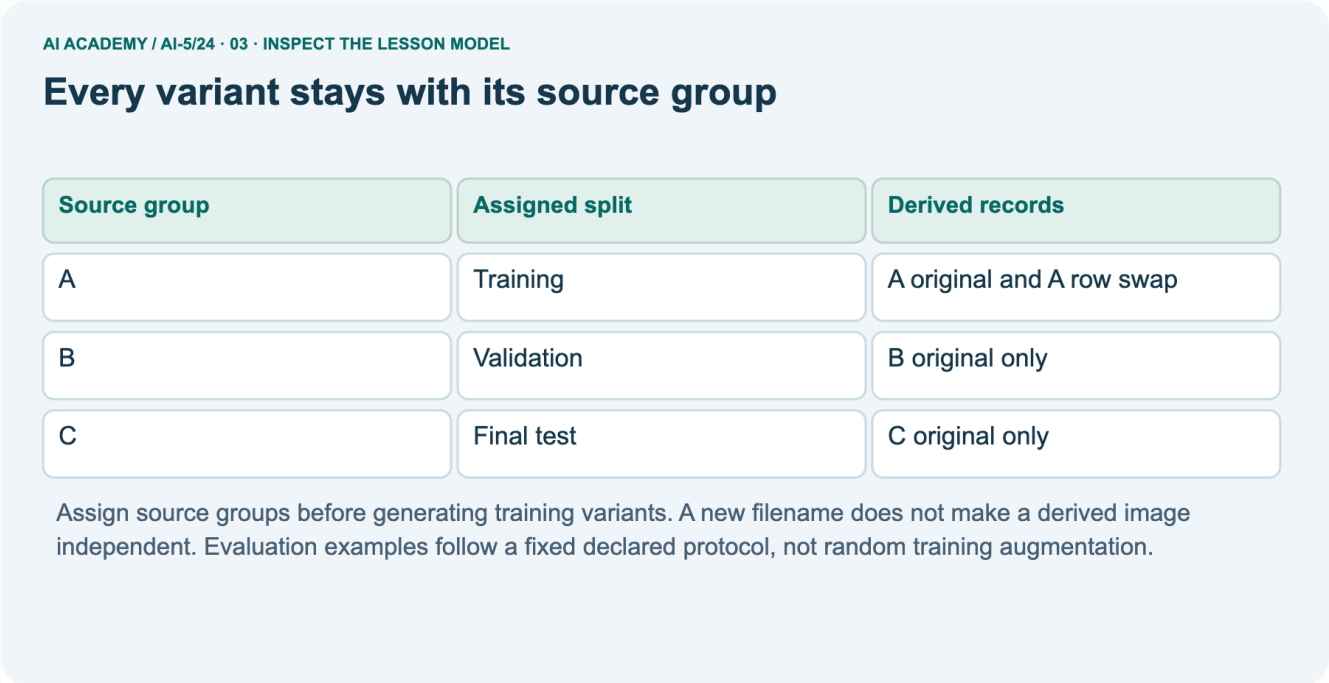
> Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if

the image is difficult to see.

03. Every variant stays with its source group



Follow the labelled picture with the matching explanation. Open the vector diagram

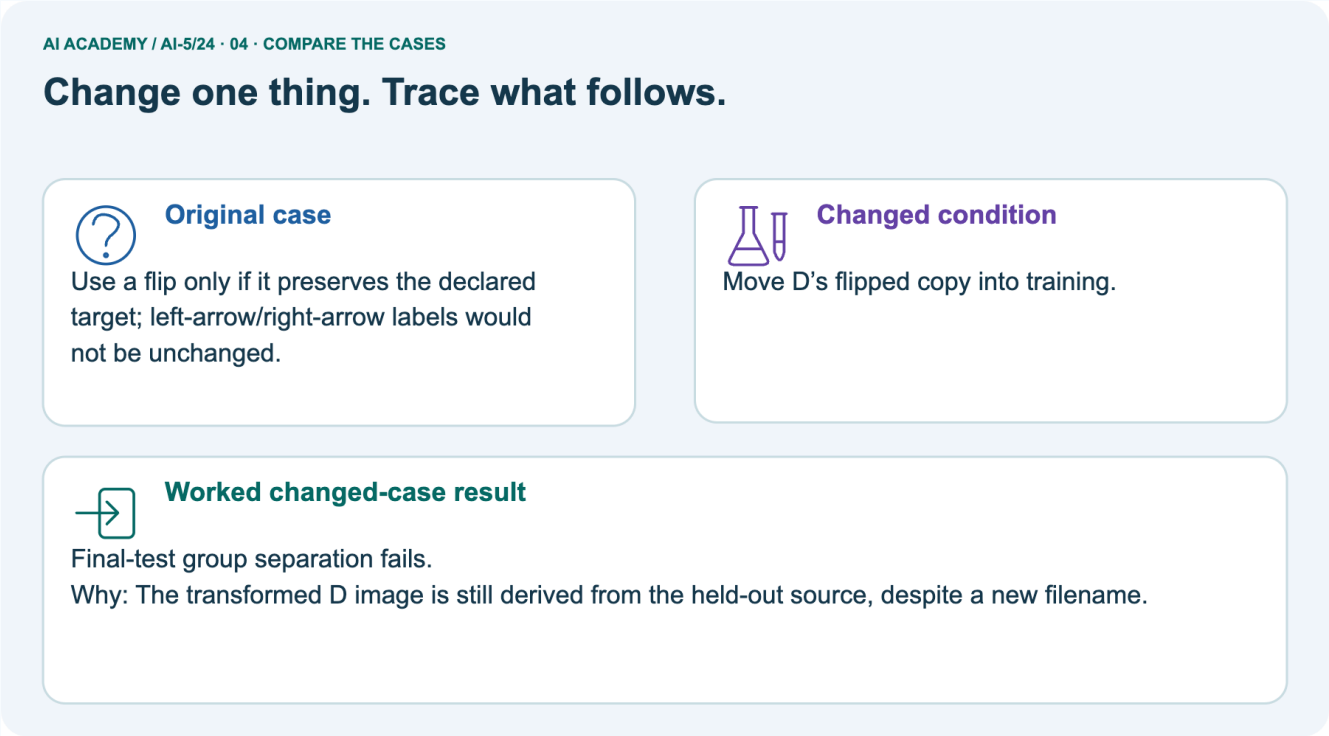
Every variant stays with its **source group**. A | **Training** | A original and A row swap. B | Validation | B original only. C | Final test | C original only

Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Use a flip only if it preserves the declared target; left-arrow/right-arrow labels would not be unchanged. Changed condition: Move D's flipped copy into **training**.

Expected result: Final-test group separation fails. Reason: The transformed D image is still derived from the held-out source, despite a new filename.

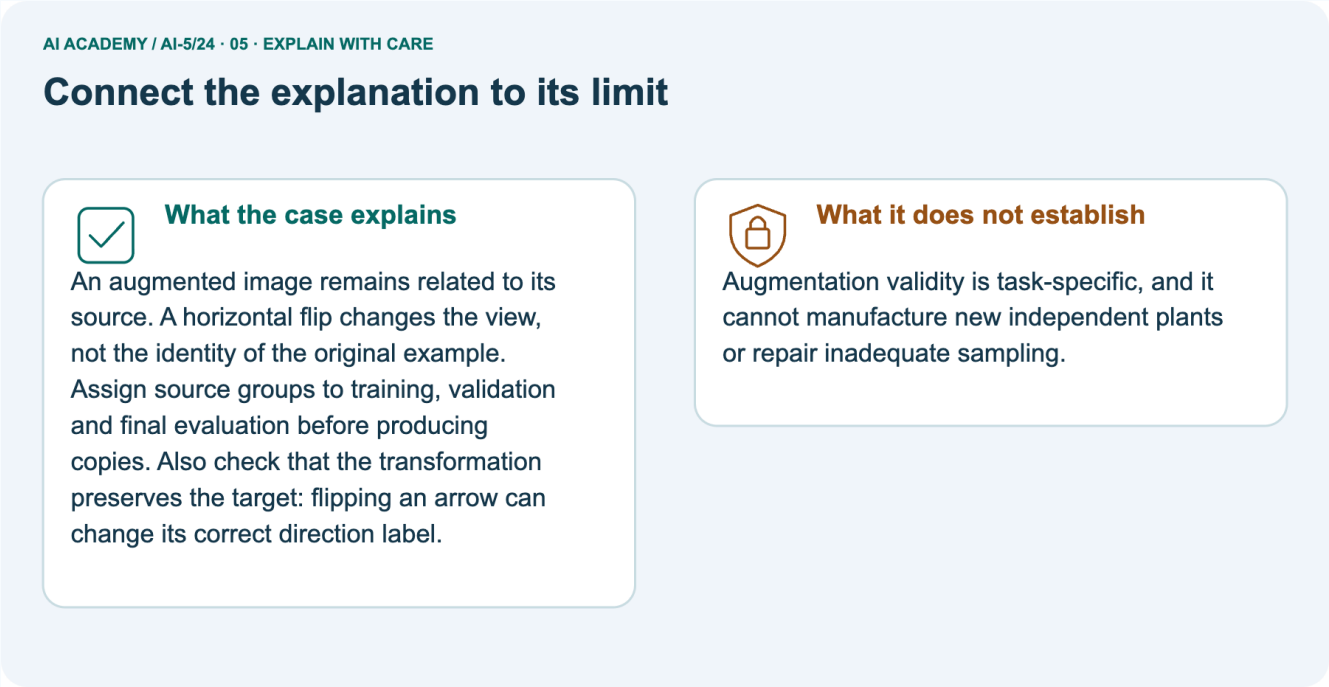
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The transformed D image is still derived from the held-out source, despite a new filename.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An augmented image remains related to its source. A horizontal flip changes the view, not the identity of the original example. Assign **source group**s to **training**, validation and final evaluation before producing copies.

Also check that the transformation preserves the target: flipping an arrow can change its correct direction label.

Limit: **Augmentation** validity is task-specific, and it cannot manufacture new independent plants or repair inadequate sampling.

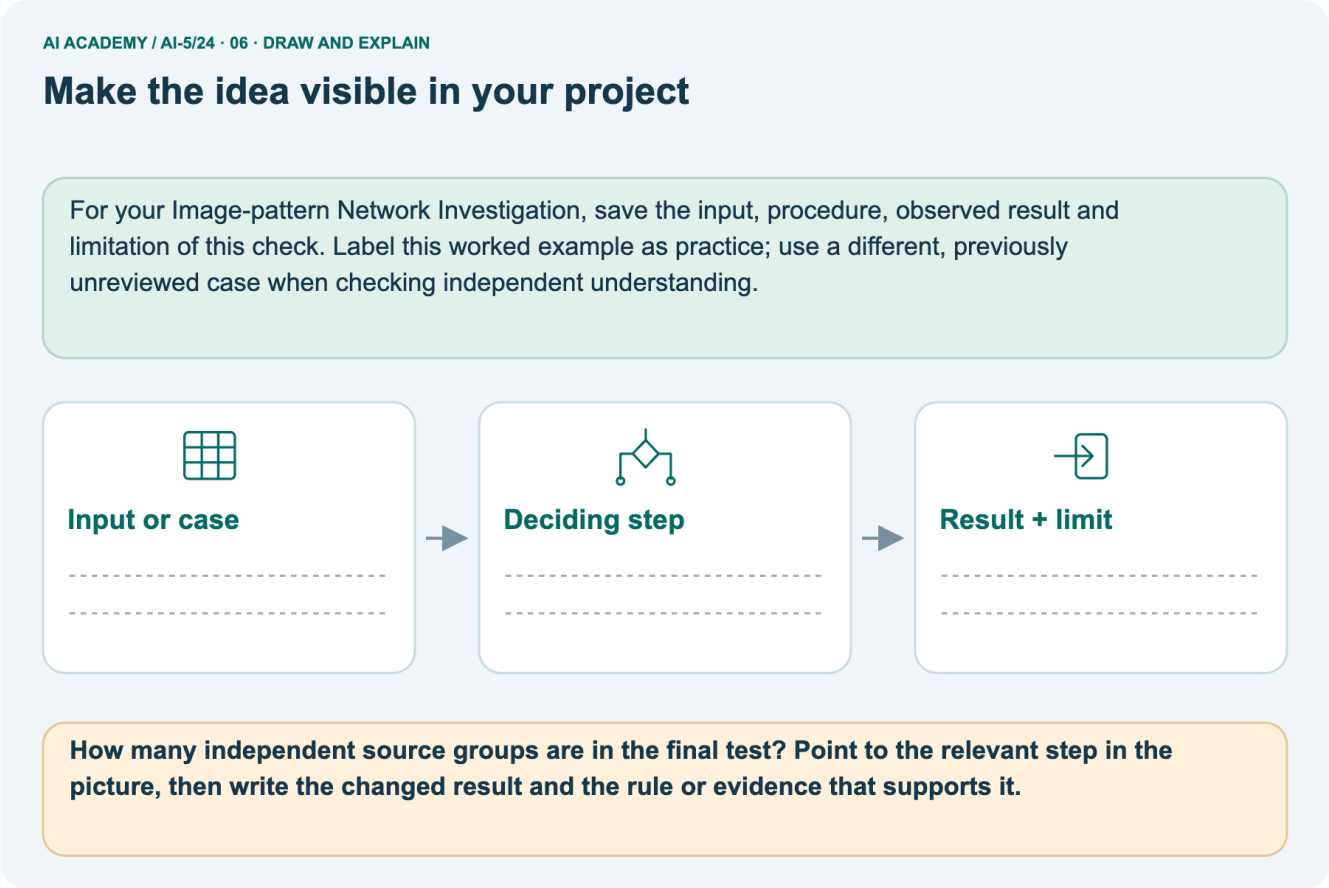
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: **Augmentation** validity is task-specific, and it cannot manufacture new independent plants or repair inadequate sampling.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: How many independent **source group**s are in the final test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: One, D, with two related records. Count both **source group**s and image rows; never describe its flip as a new independent source. Changed-case check: Final-test group separation fails.

The transformed D image is still derived from the held-out source, despite a new filename.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.